

PMLZA 6. CRC Screening (and PGMs)

UoP

David Rios Insua

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ICMAT-CSIC

- Colorectal cancer screening. Prelims
- Bayesian networks for CRC risk mapping
- Influence diagrams for CRC screening
- Multi-agent influence diagrams to incentivise CRC screening
- Discussion

Colorectal cancer screening prelims

The issue. Colorectal cancer (CRC)

- Third most common cancer
- 2020. 1.9M new cases and 930000 associated deaths globally
- 2015. EU estimated annual costs 19B€
- Early-onset rising
- Early detection entails much less costs and deaths
- Only 14% susceptible EU citizens participate in screening programs
- Current scheme. Invite people 50-70. FIT; if positive, colonoscopy

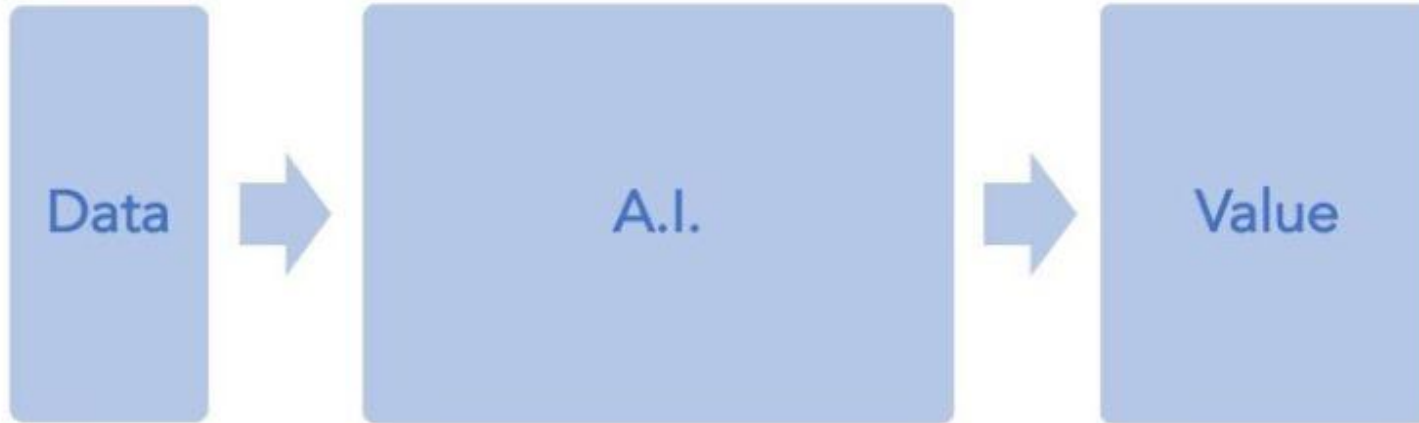
The issue. Colorectal cancer (CRC)

- New screening devices: Accurate, non-invasive, cost effective
- Raise further awareness
- More personalised screening
- Better screening resource allocation
- Incentives to nudge adherence to screening

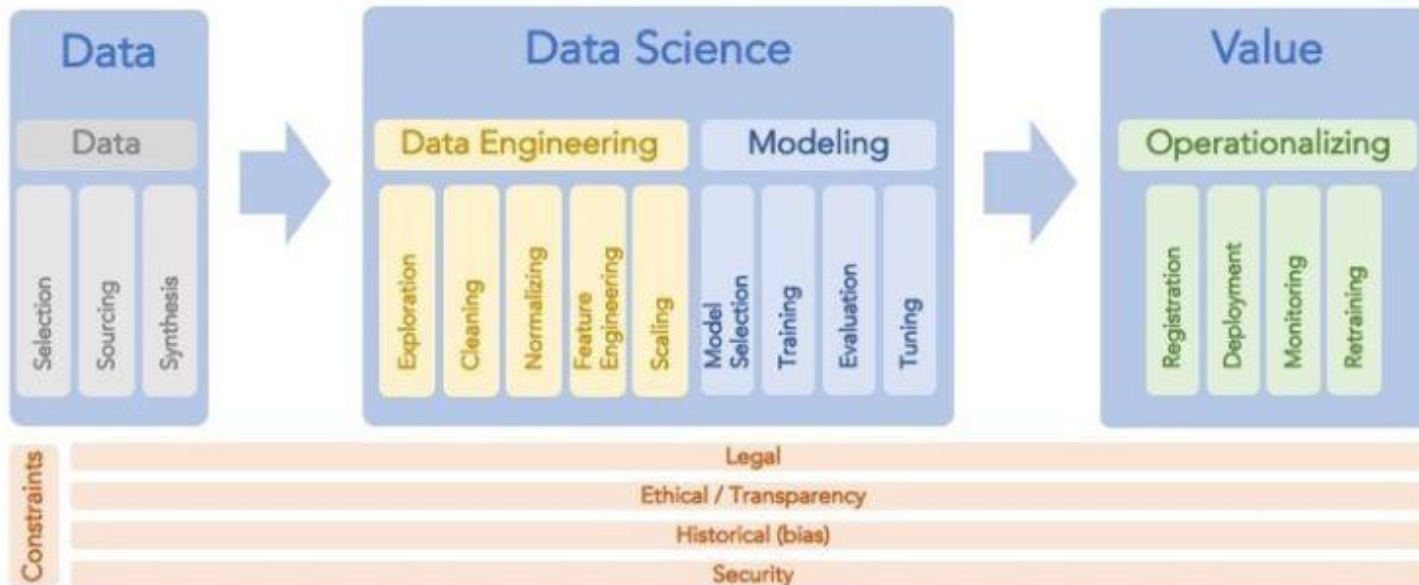
What companies think A.I. looks like



What companies think A.I. looks like

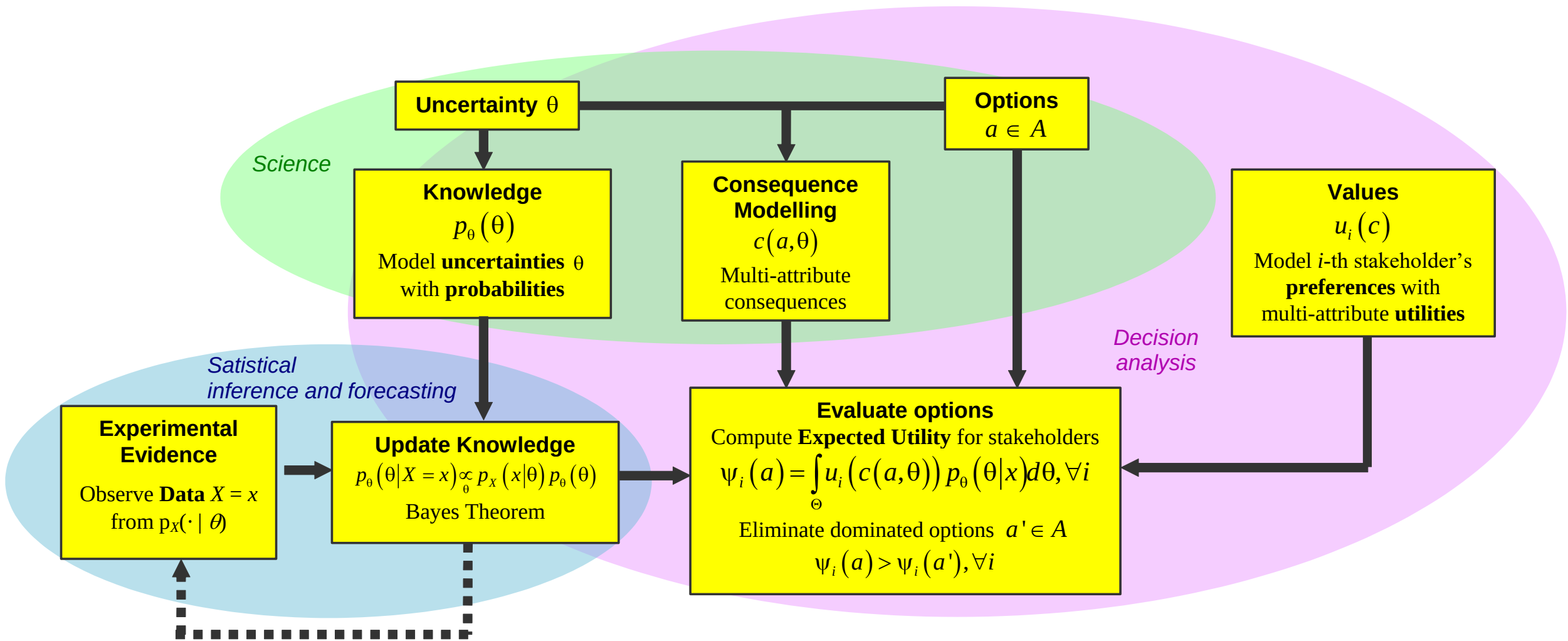


What it actually is



Dataset

- Quirónsalud. 2012-2016 Annual health reviews
- Enriched with INE data (Socioecon, Educational) Earlier
- Cleaned and consolidated (misrecorded, outliers, duplicates, missing values,...) Earlier
- 2M+ records. 66 variables



Bayesian networks for CRC risk mapping

Motivation

- Simple way to visualize structure of probabilistic models
- Designing and motivating new models
- Understanding properties like conditional independence
- Complex computations viewed through simple graphical manipulations
- Explainable and interpretable. Easy to communicate
- Classification, generation
- Deep belief nets in deep learning....

Probabilistic diagrams

- As basic tools for qualitative modelling of uncertainty use probabilistic influence diagrams a.k.a. causal networks, Bayesian networks, Belief networks,... See the excellent

http://en.wikipedia.org/wiki/Bayesian_network

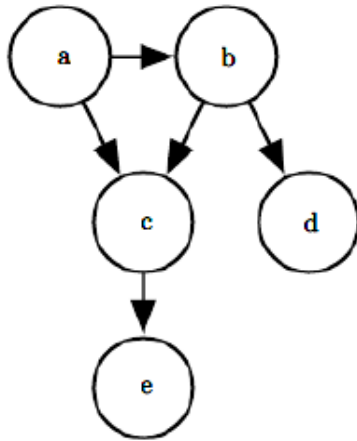
They are **influence diagrams** with chance nodes only. Qualitatively they describe a probabilistic model through

$$P(A_1, A_2, \dots, A_n) = P(A_1 \mid \text{ant}(A_1)) \dots P(A_n \mid \text{ant}(A_n))$$

where $\text{ant}(A_i)$ are the antecessors of node A_i .

Concept

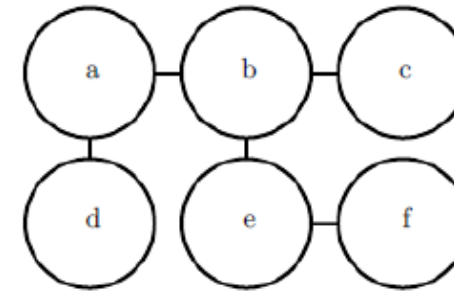
$$p(\mathbf{x}) = \prod_i p(x_i \mid \text{Pa}_{\mathcal{G}}(x_i))$$



$$p(a, b, c, d, e) = p(a)p(b \mid a)p(c \mid a, b)p(d \mid b)p(e \mid c)$$

Bayesian networks. Directed, Acyclic

$$\tilde{p}(\mathbf{x}) = \prod_{\mathcal{C} \in \mathcal{G}} \phi(\mathcal{C})$$



$$p(a, b, c, d, e, f) = \frac{1}{Z} \phi_{a,b}(a, b) \phi_{b,c}(b, c) \phi_{a,d}(a, d) \phi_{b,e}(b, e) \phi_{c,f}(c, f) \phi_{d,e}(d, e) \phi_{e,f}(e, f)$$

Markov fields. Undirected

Probabilistic diagrams with two nodes

Model for $P(A,B)$



$$P(A)P(B)$$



$$P(A) P(B|A)$$

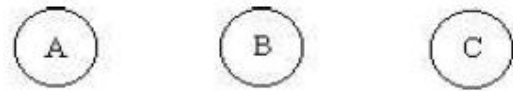


$$P(B) P(A|B)$$

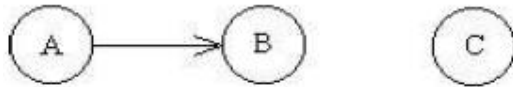
First case, A and B are independent. We move from second to third, and viceversa, via Bayes formula

Probabilistic diagrams with three nodes

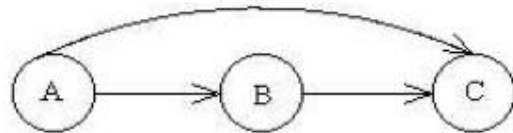
Model $P(A, B, C)$



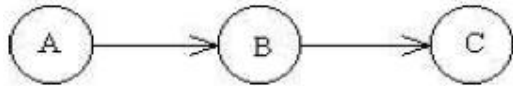
$P(A)P(B)P(C)$



$P(A) P(B|A) P(C)$



$P(A)P(B|A)P(C|A,B)$

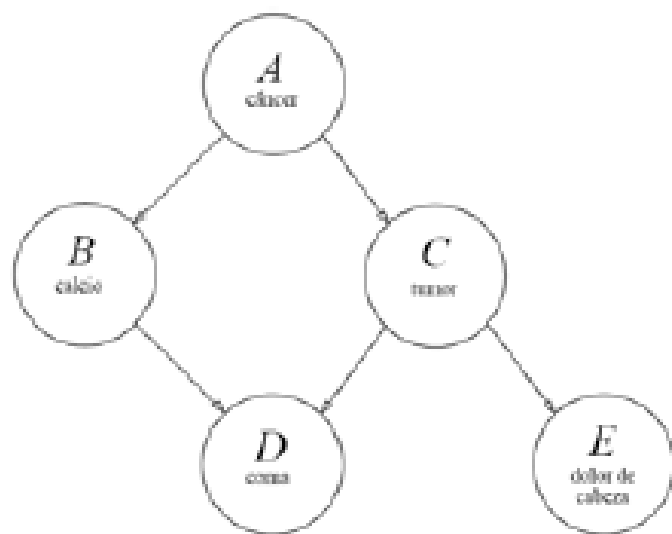


$P(A)P(B|A)P(C|B)$

First case, independence. Fourth case, A and C are conditionally independent given B.

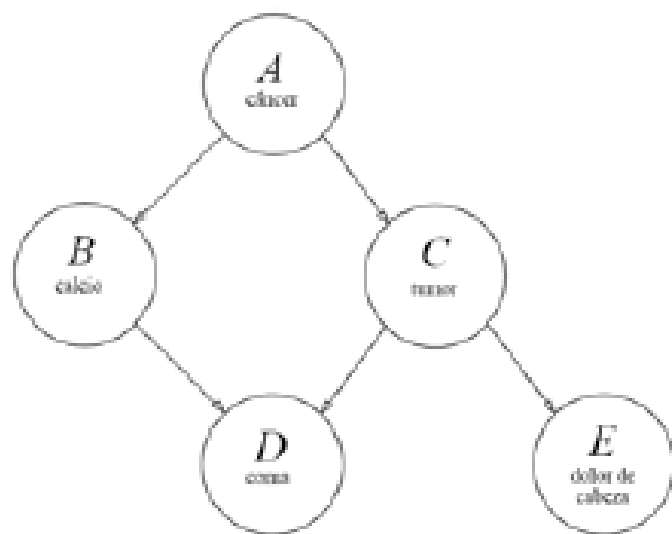
Read http://en.wikipedia.org/wiki/Conditional_independence

The hidden info



$$P(A, B, C, D, E) = P(A)P(B|A)P(C|A)P(D|B, C)P(E|C)$$

The hidden info



a	0.2
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	a	$\bar{a}$
c	0.2	0.05

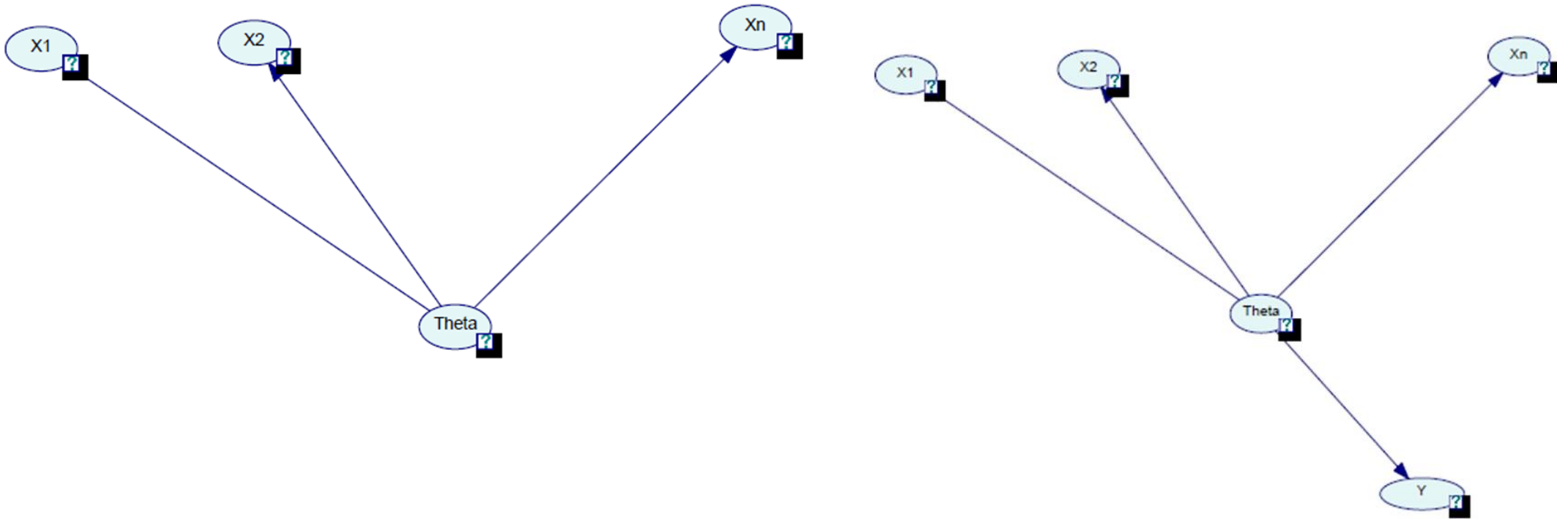
	b, c	$\bar{b}, c$	$b, \bar{c}$	$\bar{b}, \bar{c}$
d	0.8	0.8	0.8	0.05

	a	$\bar{a}$
b	0.8	0.2

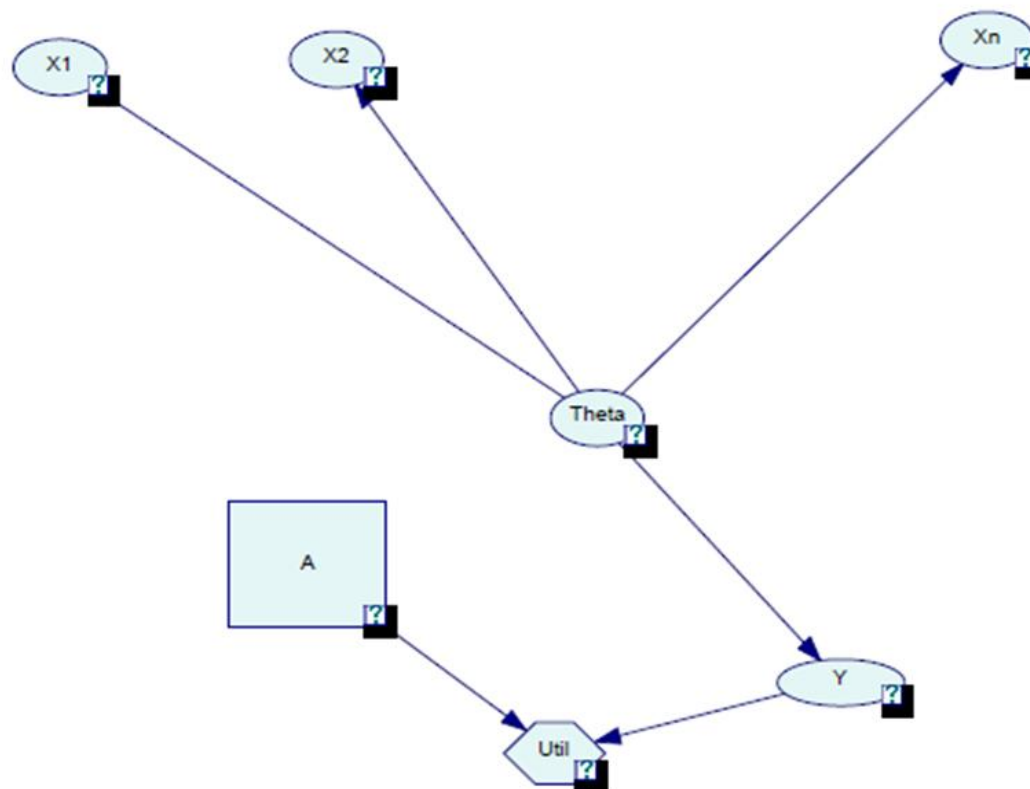
	c	$\bar{c}$
e	0.8	0.6

$$P(A, B, C, D, E) = P(A)P(B|A)P(C|A)P(D|B, C)P(E|C)$$

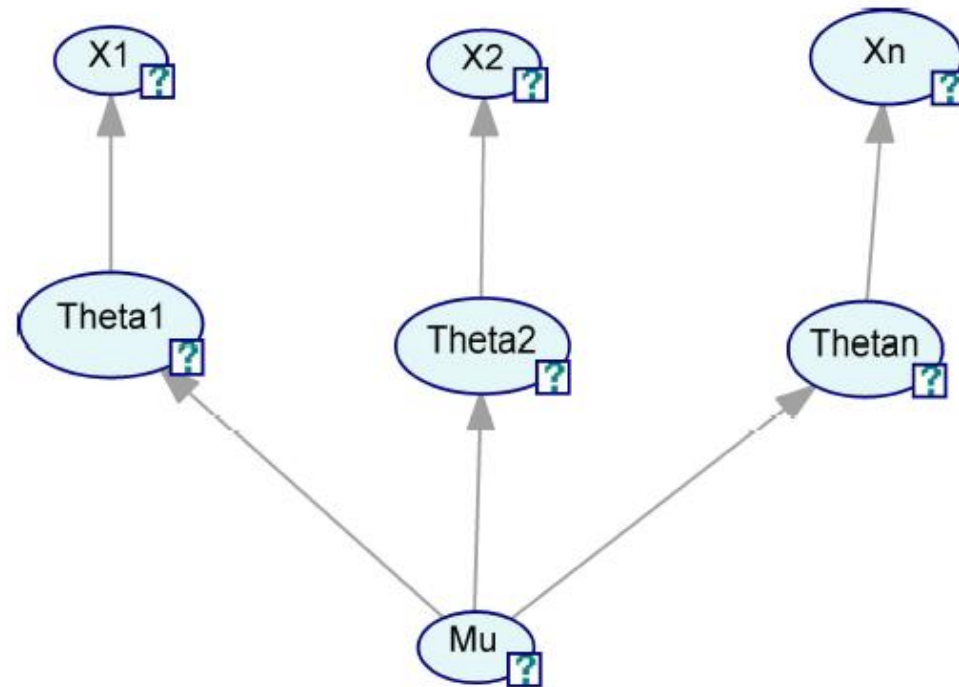
ML models as PGMs. Inference and Prediction



ML models as PGMs. Decision Analysis

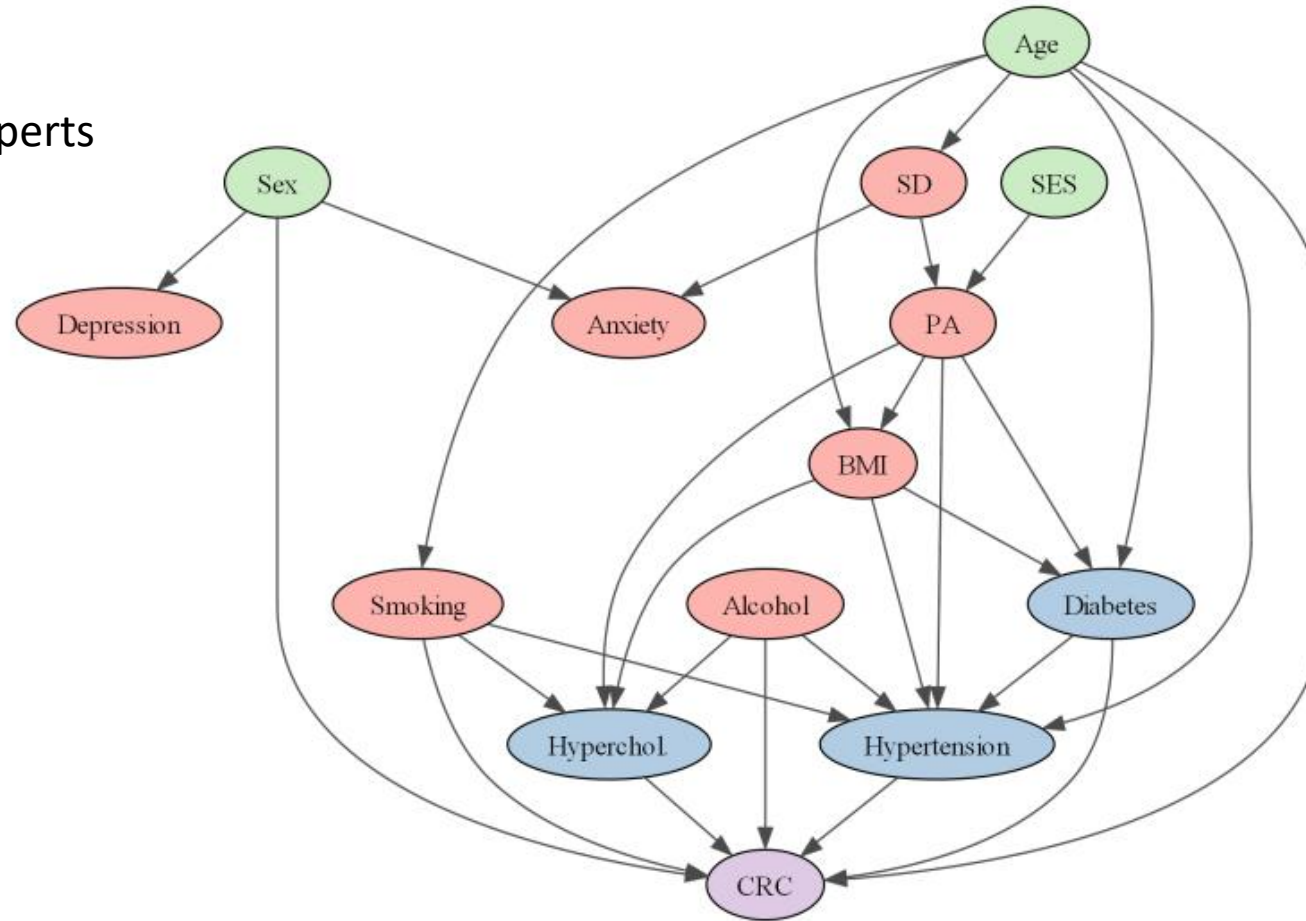


ML models as PGMs. Hierarchical models



Through standard
bibsearch (adapted to
database)

Then reviewed by experts

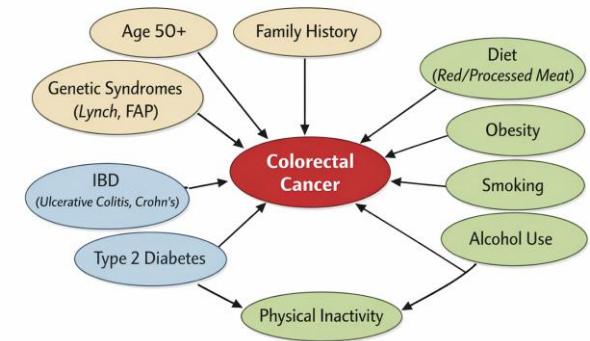


Non-modifiable
Modifiable
Medical conditions
CRC

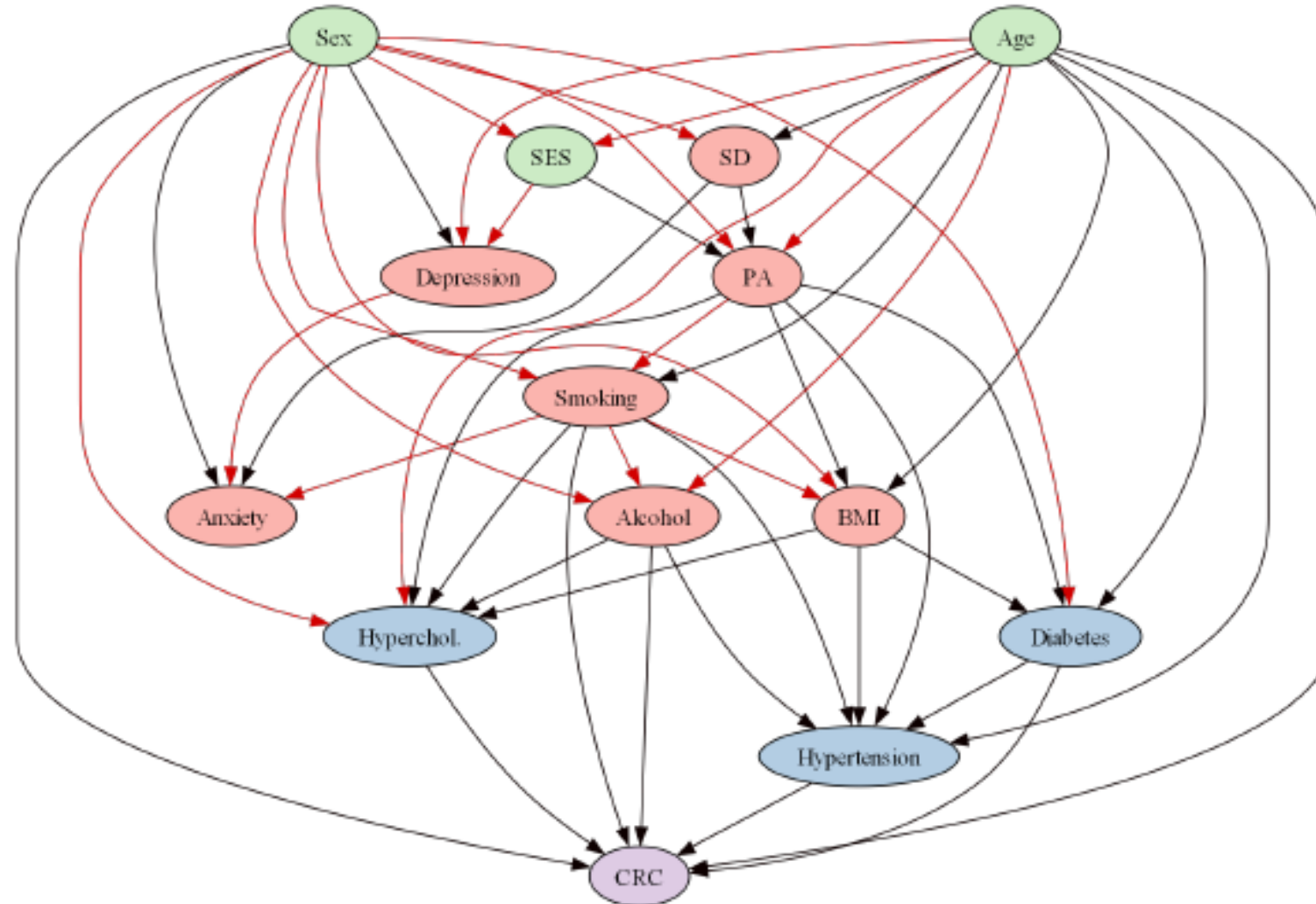
From ChatGPT....

- Colorectal cancer risk comes from a mix of **age, genetics, lifestyle, and certain medical conditions**. Here are the big ones, grouped so they're easier to digest:

- Age & Genetics**
 - Age 50+** (risk rises steadily with age, though rates are increasing in younger adults)
 - Family history** of colorectal cancer or advanced polyps
 - Inherited syndromes**, like:
 - Lynch syndrome
 - Familial adenomatous polyposis (FAP)
- Personal Medical History**
 - Previous colorectal cancer or adenomatous polyps**
 - Inflammatory bowel disease** (ulcerative colitis or Crohn's disease involving the colon)
 - Type 2 diabetes**
- Lifestyle Factors**
 - Diets high in red or processed meats**
 - Low fiber, fruit, and vegetable intake**
 - Physical inactivity**
 - Overweight or obesity**
 - Smoking**
 - Heavy alcohol use**
- Other Factors**
 - Male sex** (slightly higher risk than females)
 - African American and Ashkenazi Jewish ancestry** (higher incidence and mortality, for complex genetic and social reasons)



$$\begin{aligned}
 p(v_{sex}, \dots, v_{depression}) = & \left[p(v_{sex})p(v_{age})p(v_{SES}|v_{sex}, v_{age}) \right] \times \\
 & \left[p(v_{SD}|v_{sex}, v_{age})p(v_{PA}|v_{sex}, v_{age}, v_{SD}, v_{SES})p(v_{depr}|v_{sex}, v_{age}, v_{SES}) \right. \\
 & p(v_{smok}|v_{sex}, v_{age}, v_{PA})p(v_{alc}|v_{sex}, v_{age}, v_{smok}) \\
 & \left. p(v_{BMI}|v_{sex}, v_{age}, v_{PA}, v_{smok})p(v_{anx}|v_{sex}, v_{SD}, v_{smok}, v_{depr}) \right] \times \\
 & \left[p(v_{hypchol}|v_{sex}, v_{age}, v_{PA}, v_{smok}, v_{BMI}, v_{alc})p(v_{diab}|v_{sex}, v_{age}, v_{PA}, v_{BMI}) \right. \\
 & \left. p(v_{hypten}|v_{age}, v_{PA}, v_{smok}, v_{BMI}, v_{alc}, v_{diab}) \right] \times \\
 & p(v_{CRC}|v_{sex}, v_{age}, v_{alc}, v_{smok}, v_{hypchol}, v_{hypten}, v_{diab})
 \end{aligned}
 \tag{1}$$

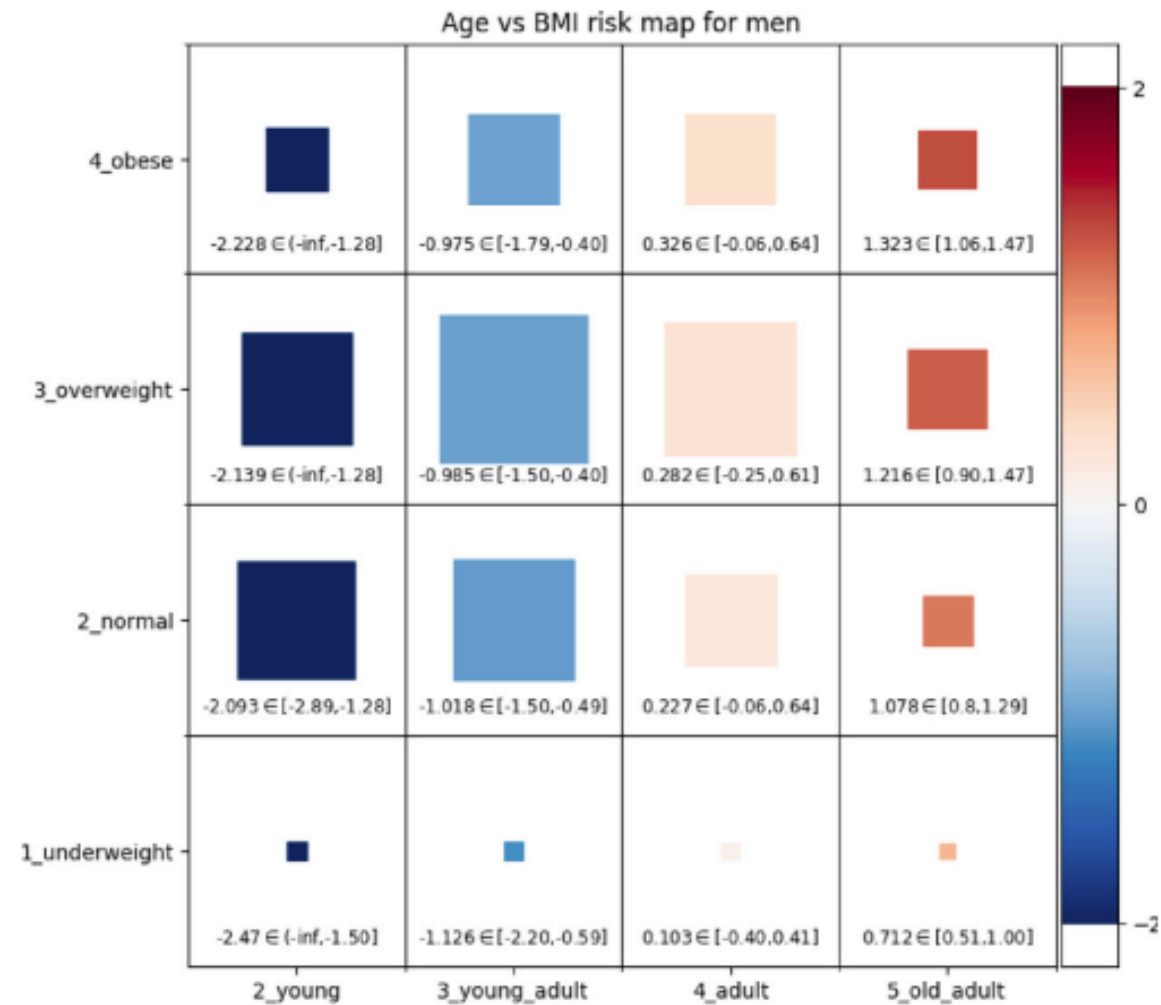


Probability tables

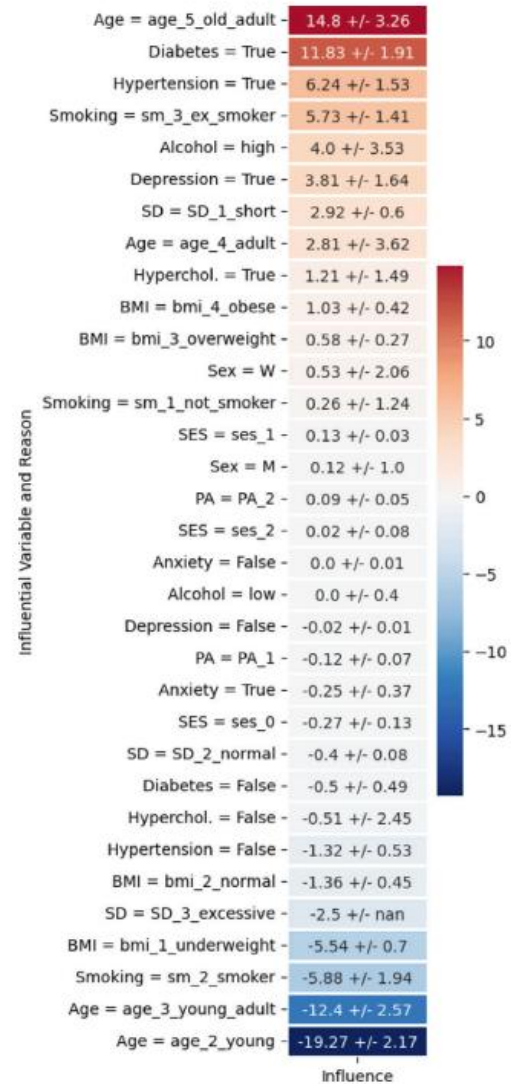
Priors

Validation

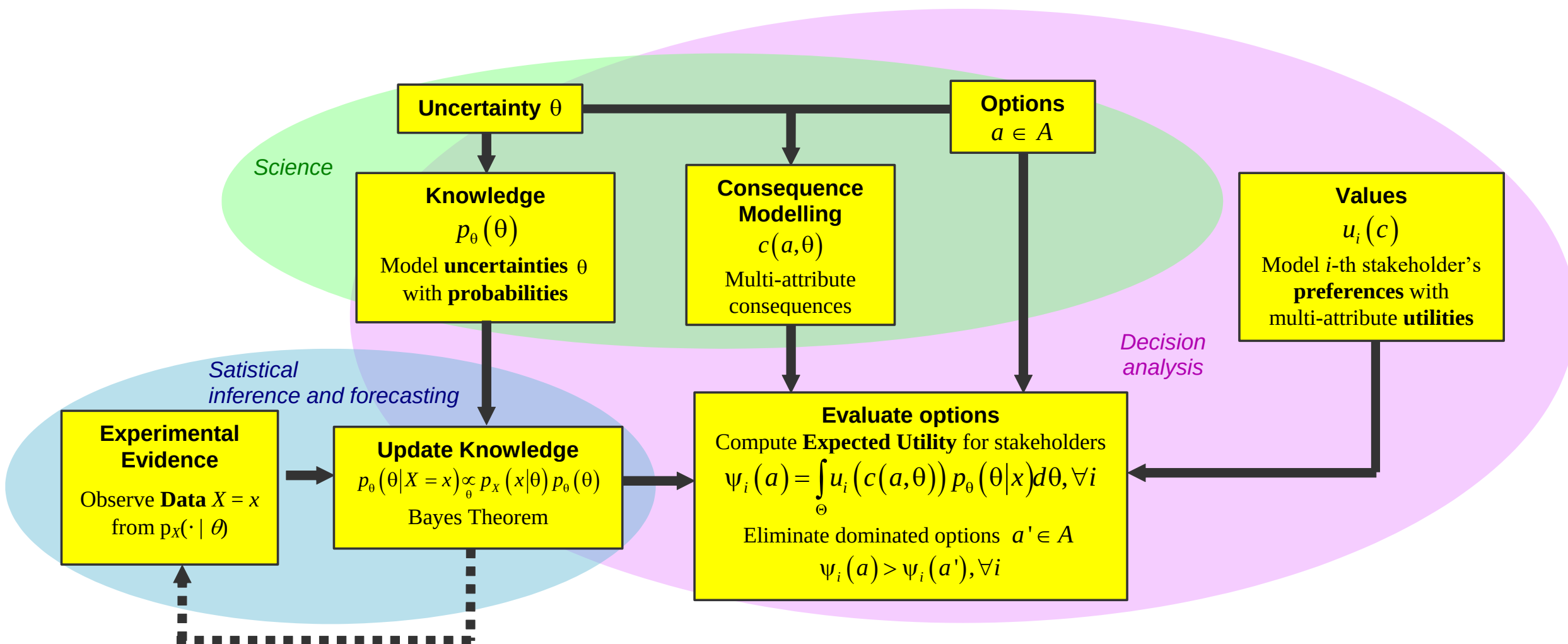
Risk maps



Influential variables



Influence diagrams for CRC screening

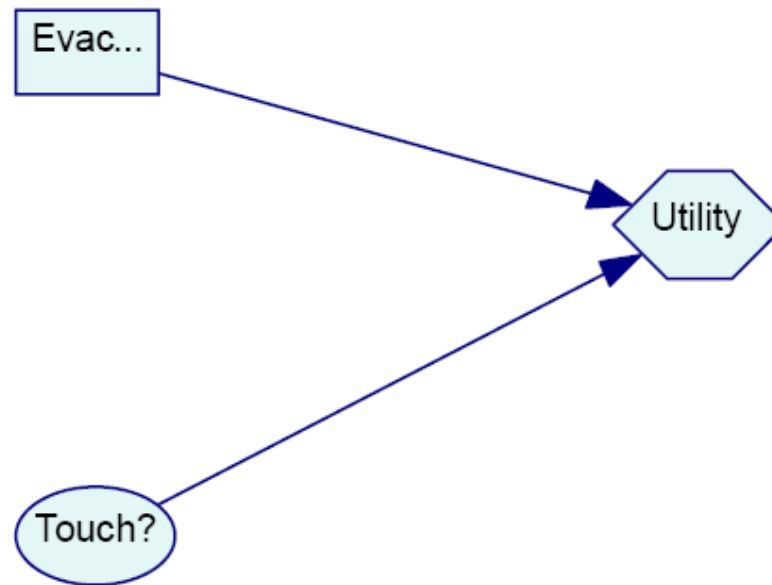


Influence Diagrams

- Tool to structure (and solve) decision making problems
- Direct acyclic graph $G=(N,A)$
- Three main types of nodes.
 - Chance. Circle
 - Decision. Square
 - Value. Hexagon, Diamond
 - Fourth type of node. Deterministic. Double circle
- Two types of arcs
 - Arcs into decision nodes
 - Arcs into chance and value nodes

Suppose you're ComValencia's president. There is a major flood threat. Would you issue an evacuation order?

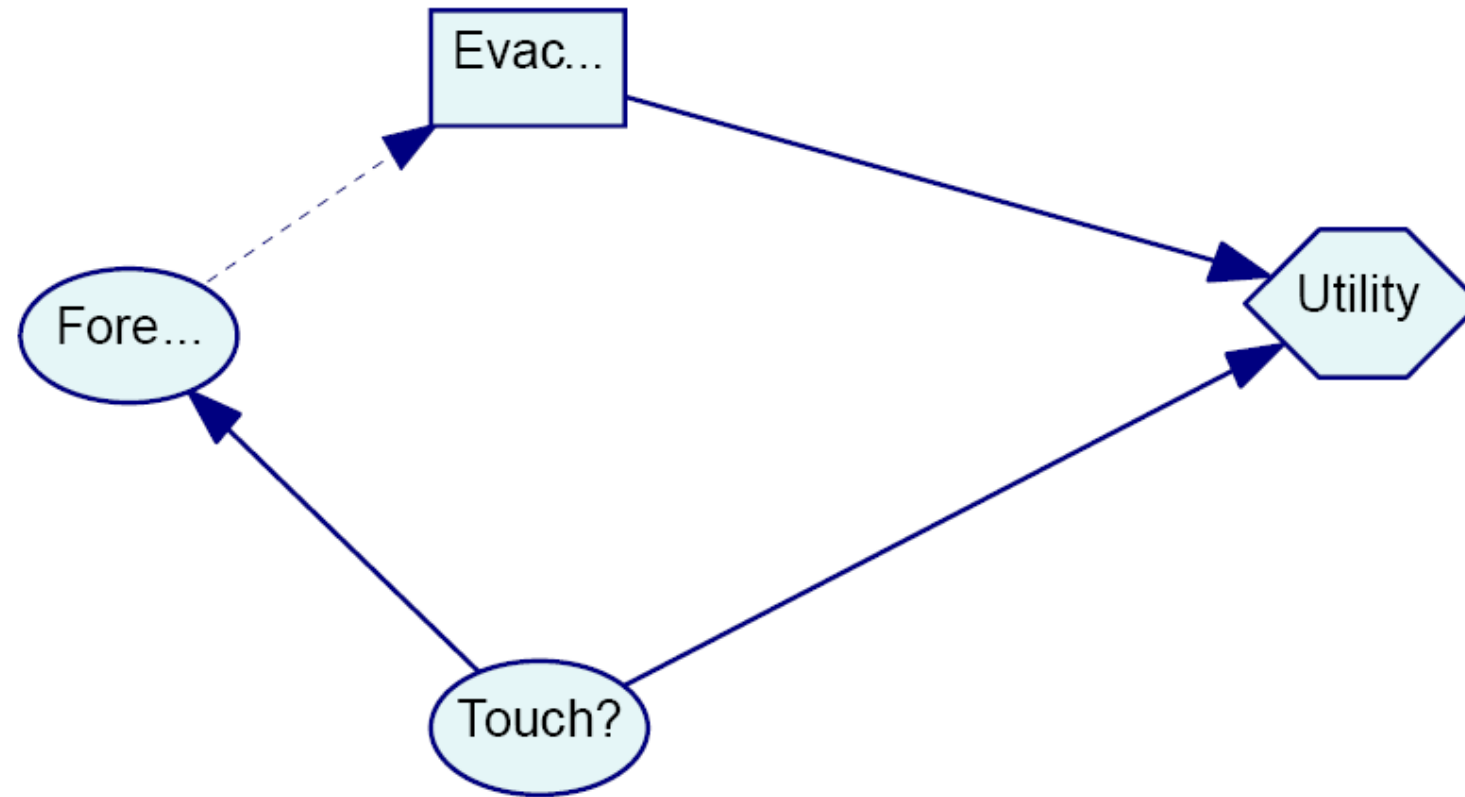
Decision making under risk



Suppose you're ComValencia's president. There is a major flood threat. Would you issue an evacuation order?

You have as info a forecast from the AEMET. But the forecast is not perfect...

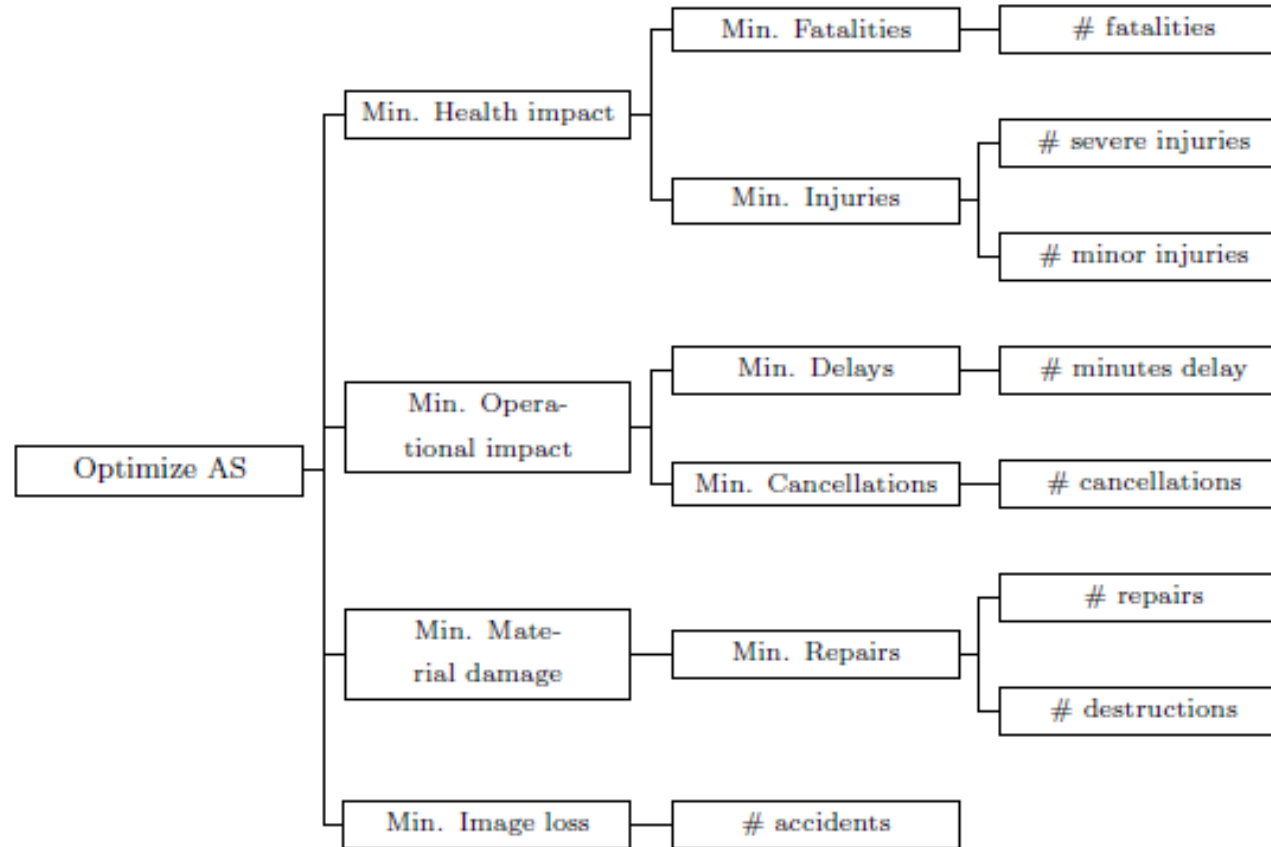
Decision under risk with imperfect information



Multiple objectives

- Hierarchy
- From top level generic objective
- To more precise subobjectives
- Until low subobjectives determined.
 - Relevant
 - Non redundant
 - Measurable
- Natural, Constructed, Proxy

Multiple objectives in aviation safety. AESA



Modeling preferences under uncertainty

Build a function associating a number (utility) to each consequence, so that we compute the expected utility of each action

$$u(a) = \sum_{\theta} u(a, \theta) P(\theta)$$

Optimal alternative: **maximum expected alternative**

Modeling preferences under uncertainty

Which of these lotteries do you prefer?

$$\mathbf{A} = \begin{pmatrix} 1/2 & 1/2 \\ 1000 & -600 \end{pmatrix} \text{ } y \text{ la } \mathbf{B} = \begin{pmatrix} 1/2 & 1/2 \\ 100000 & -50000 \end{pmatrix}$$

Modeling preferences under uncertainty

DM expresses preferences over lotteries

- Preferences rational: transitive and complete
- Independence property
- Continuity property (no heaven, no hell)

Preferences modeled through expected utility

Utility is affine unique

$$\max_a \int u(c(a, \theta)) p(\theta | x) d\theta$$

Modelling preferences. Risk attitudes

- Three basic risk attitudes
 - Risk aversion. Concave utility
 - Risk proneness. Convex utility
 - Risk neutrality. Linear utility

Consequence varying risk attitudes

Risk attitudes

Compare lotteries A and B. Which one do you prefer?

$$\mathbf{A} = \begin{pmatrix} 1/2 & 1/2 \\ 0 & 100000 \end{pmatrix} \text{ y } \mathbf{B} = \begin{pmatrix} 1 \\ 50000 \end{pmatrix}.$$

Risk attitudes

Compare lotteries A and B. Which one do you prefer?

$$A = \begin{pmatrix} 1/2 & 1/2 \\ 0 & 100000 \end{pmatrix} \text{ y } B = \begin{pmatrix} 1 \\ 50000 \end{pmatrix}.$$

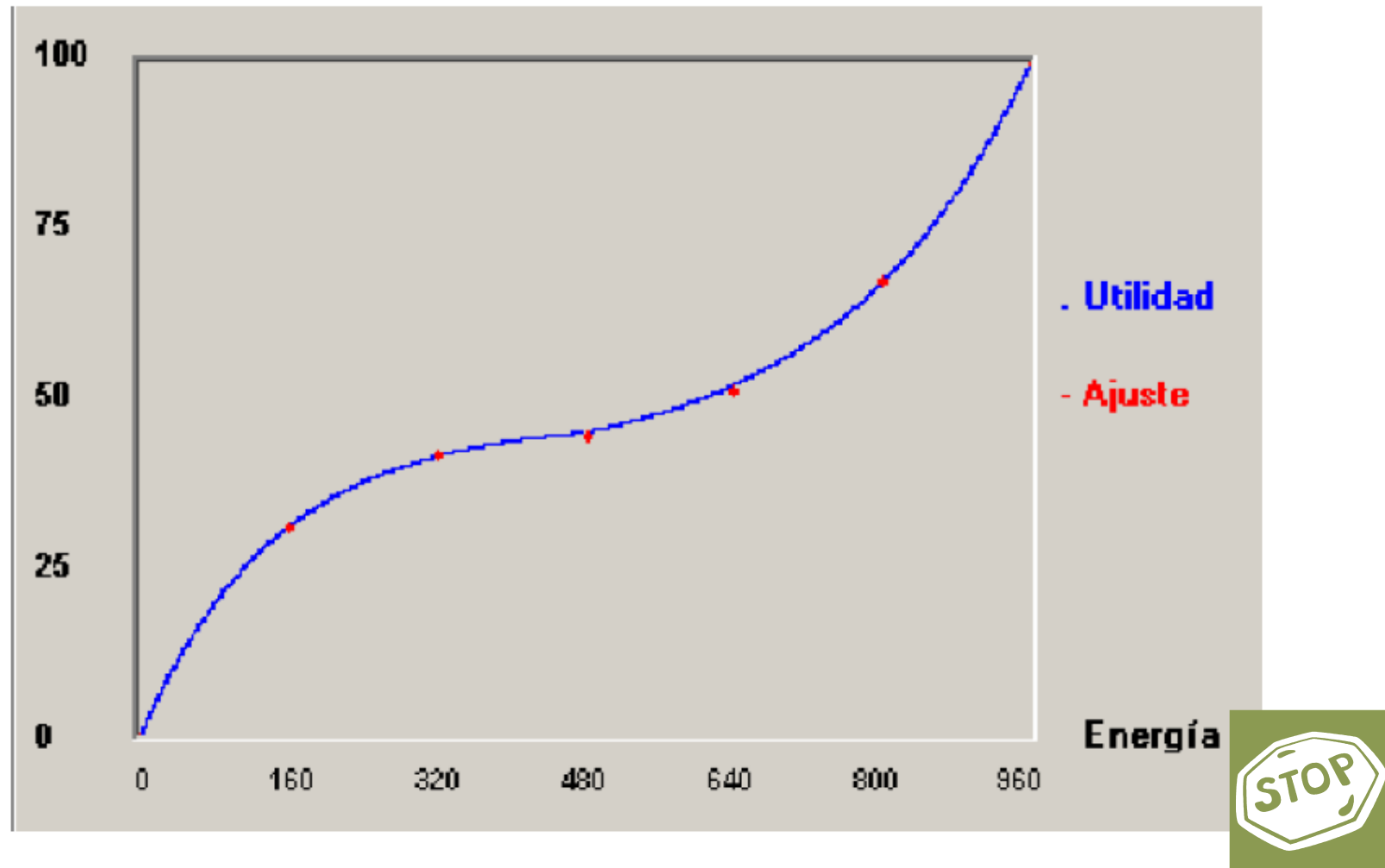
If you prefer B to A, you seem to prefer the sure prize to the lottery (risk aversion)

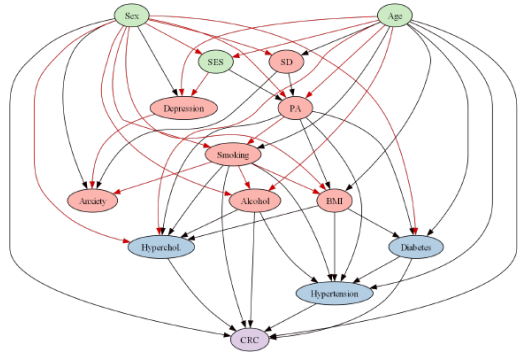
In such case, the expected utility of B should be bigger than that of A:

$$1/2 * u(0) + 1/2 * u(100000) = 1/2 < 1 * u(50000)$$

Graphically...

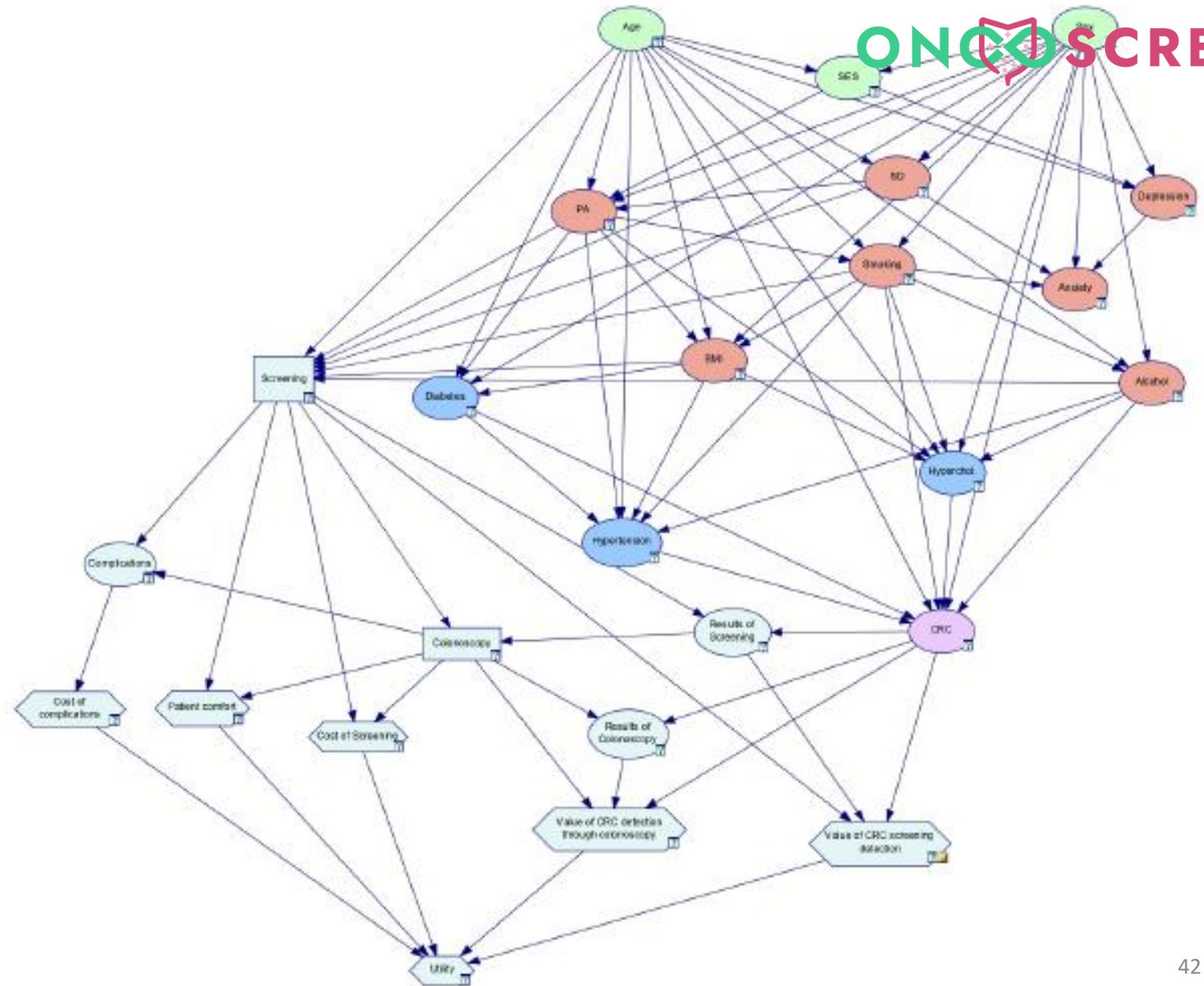
Which risk attitude??





Cost intervention
Cost complications
Comfort
Information

Multicriteria value function
Risk averse utility function

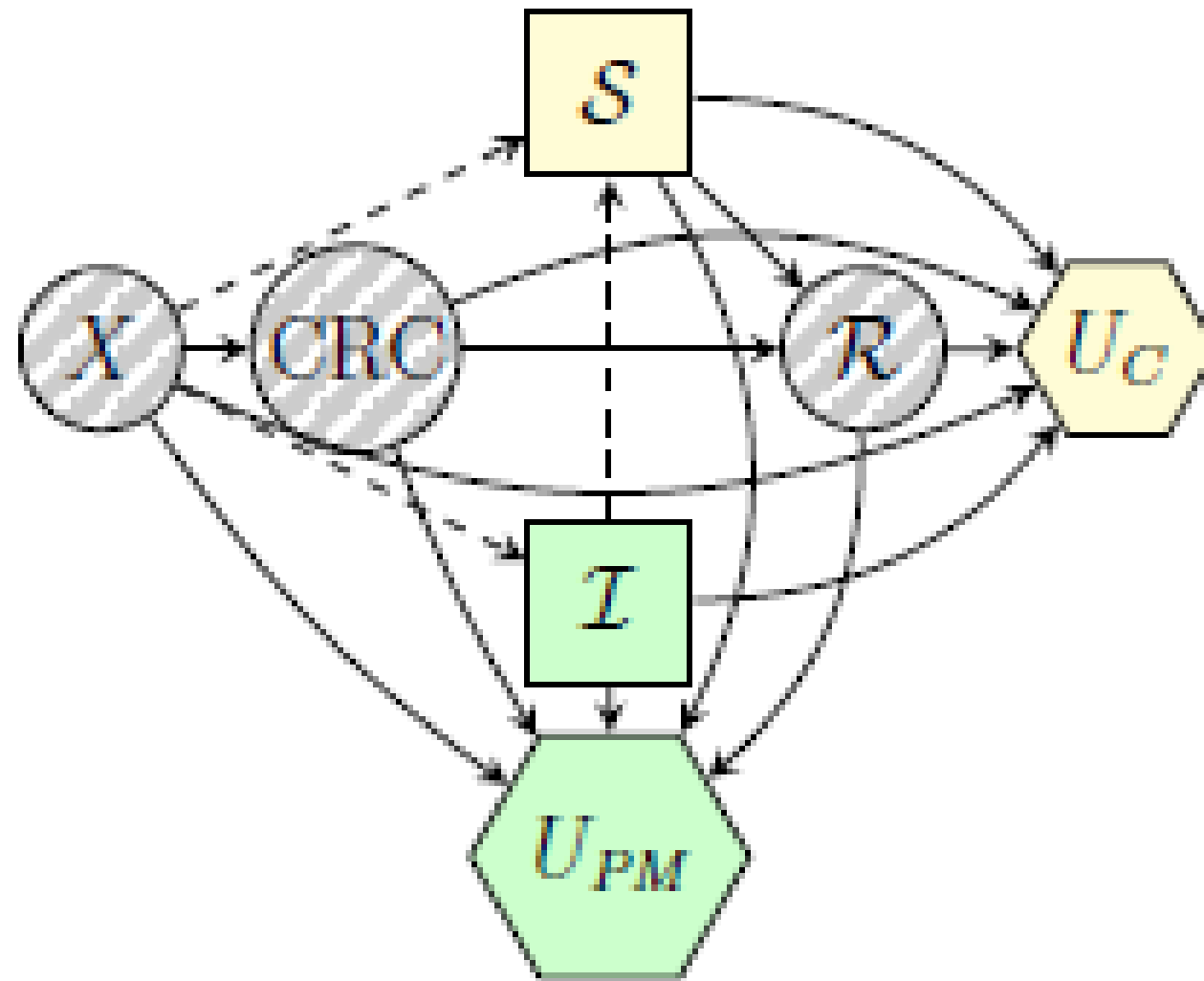


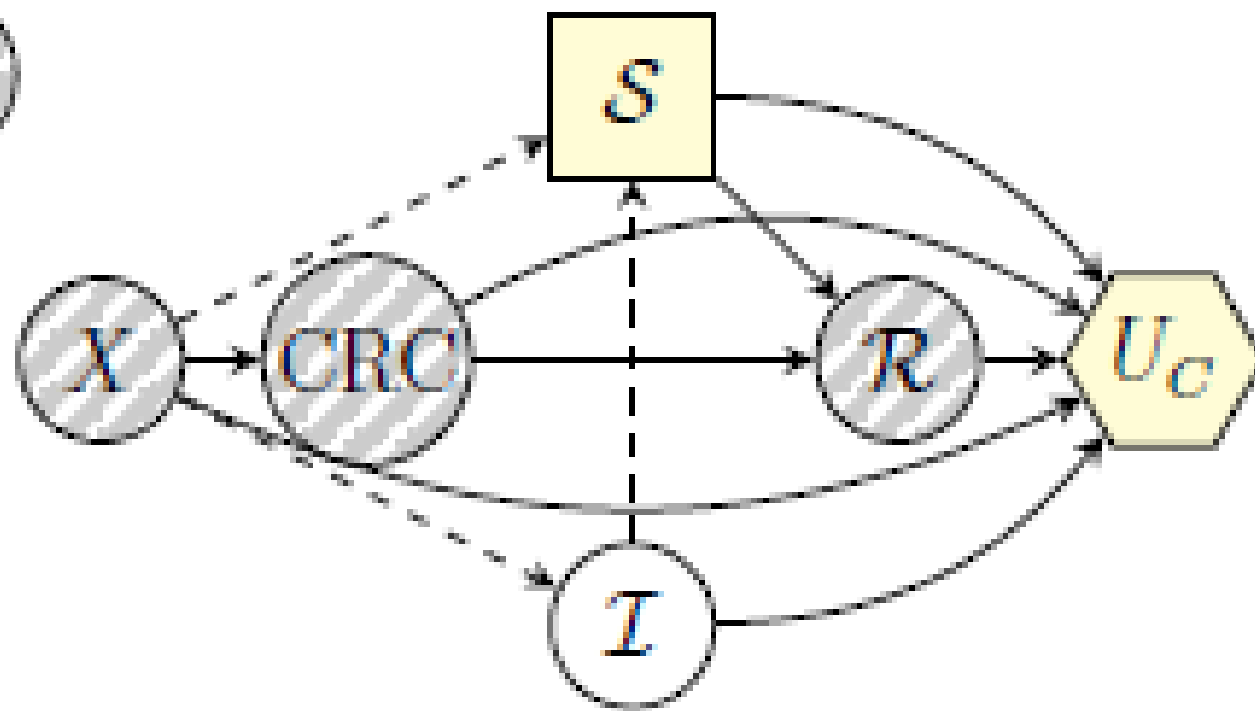
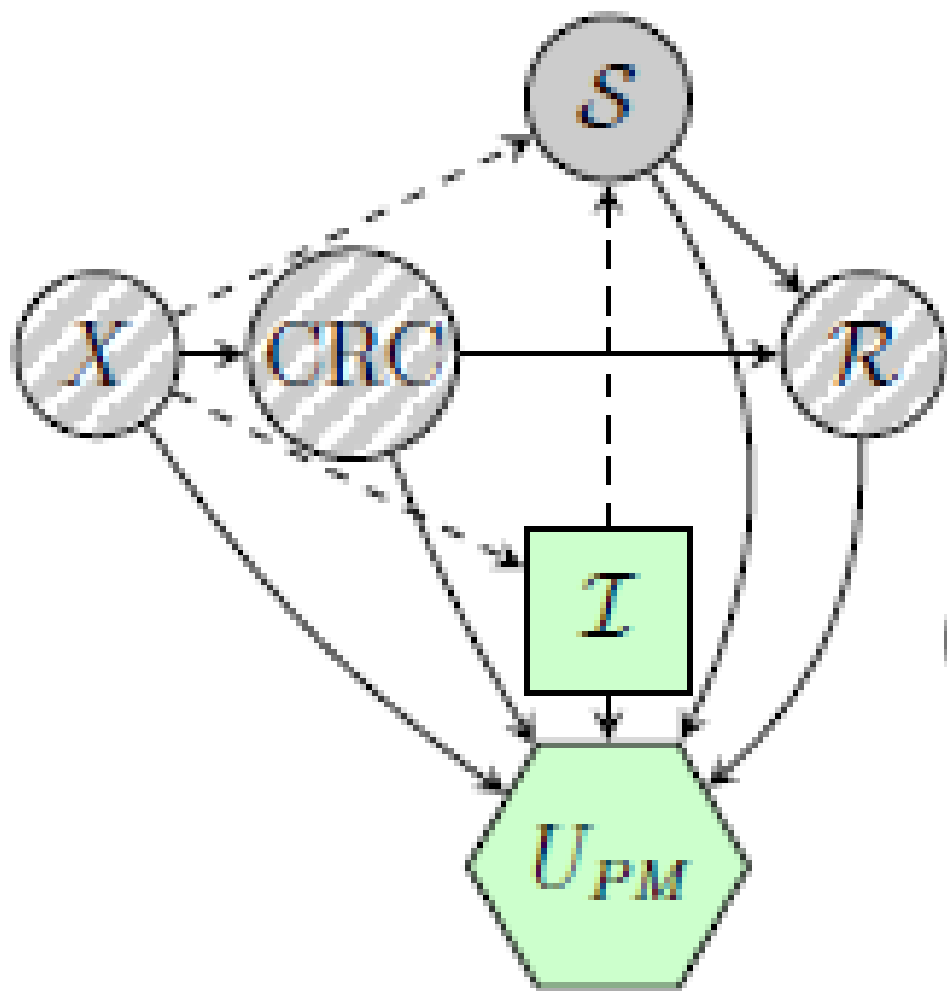
Use cases

- Individual recommendations. Personalised screening based on risk
- Benchmarking new screening devices (ONCOSCREEN)
- Assessing screening programs (and designing)

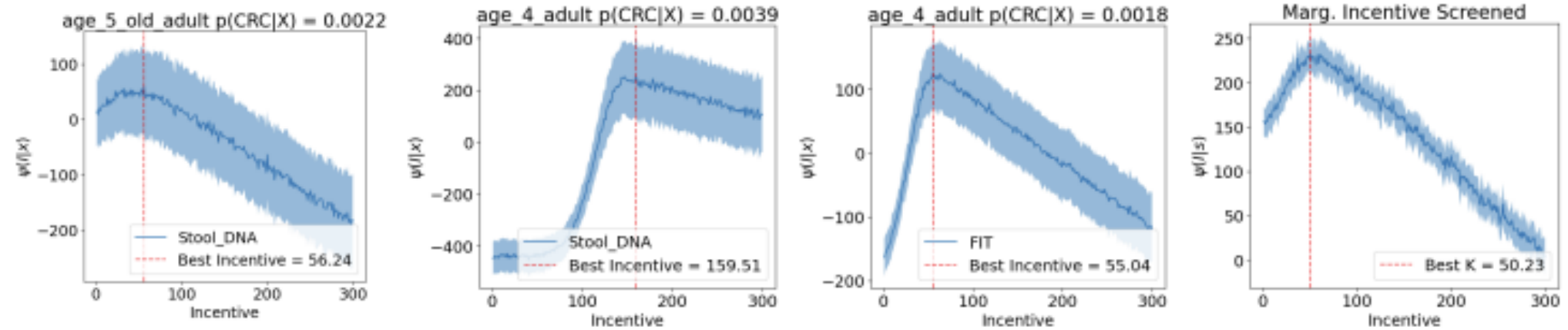
Current Spanish Strategy	Proposed Strategy
• If patient >50 years old: Send FIT invitation regardless their features. • Else: No screening.	For patient with features X: • if $p(CRC X) < \theta_1 \rightarrow$ No screening • if $\theta_1 \leq p(CRC X) < \theta_2 \rightarrow$ FIT • if $p(CRC X) \geq \theta_2 \rightarrow$ sDNA
Example: Man under 50 with high risk (due to e.g. high alcohol consumption, overweight, exsmoking) would be detected by the model as a higher risk patient than a healthy man above 60.	
Results: Extrapolating results in Spain's target population, this strategy would detect 134 patients more.	

Multi-agent IDs for incentivising CRC screening

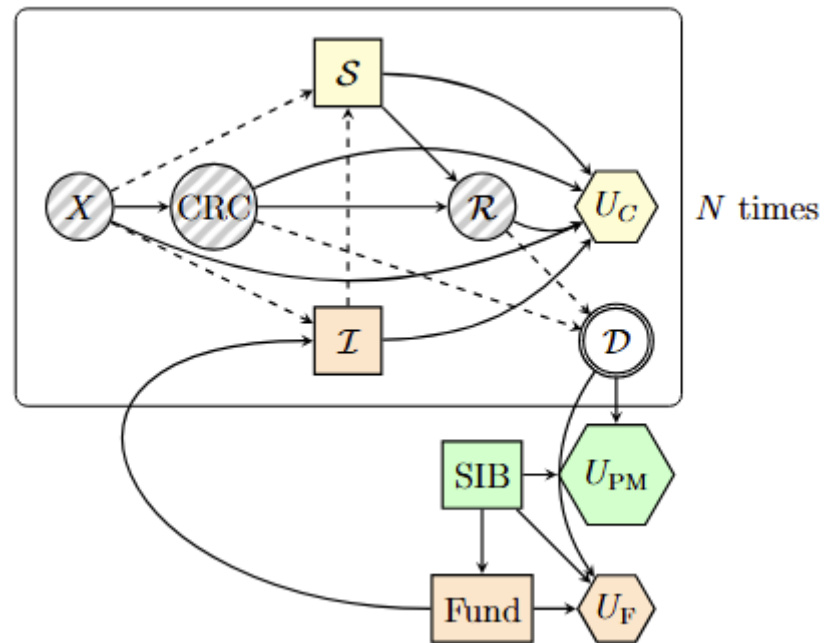




Incentives



Incentives. Social impact bond



Discussion

Bayesian networks

Influence diagrams

Multi-agent influence diagrams

A very complex case in CRC screening

AI should end in supporting/automating decisions

Benchmarking the new devices

Questions? david.rios.insua@gmail.com